Public CAD Is Not Operational Ground Truth

A Hurricane Ida Stress Test of Public Dispatch Data

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Abstract

Researchers commonly measure police response time with timestamps recorded in computer-aided dispatch (CAD) systems. When CAD records are publicly released, that use requires stable field definitions and coverage. In New Orleans, the released record changed abruptly on 28 July 2021. Before that date, none of 419,840 non-officer-initiated records contained an arrival timestamp without a dispatch timestamp. Within two days, that configuration became common, while dispatch-field coverage in officer-initiated records fell from 100 percent to 1 percent. Five weeks later, Hurricane Ida produced a temporary additional shift: among records with a valid arrival field, the public missing-dispatch share reached 43.4 percent on 31 August and 49.3 percent on 1 September. The largest half-day change was 53.2 percentage points, larger than all 151 post-change ordinary-week comparisons. These results document instability in the released measurement product, not physical police response, police performance, or the mechanism generating the public fields. Public CAD timestamps should be used as operational clocks only after field meaning, coverage, and provenance are validated.

Keywords: calls for service; police responsiveness; administrative data; Hurricane Ida; measurement error; administrative data quality
JEL codes: C18, C55, H12, K42

1 Introduction

Researchers and public agencies commonly construct police response-time measures from timestamps recorded in CAD systems. When CAD records are publicly released, that use assumes the fields retain stable meanings and coverage. This paper tests that assumption in New Orleans before treating the released fields as performance measures. It observes whether public dispatch and arrival fields are present and how their joint pattern changes. A missing dispatch timestamp may reflect a missing operational stage, but it may also reflect workflow, later entry, retention, redaction, or export logic. These field-presence measures describe the released record; they are not response times, physical-event indicators, or measures of service quality.

Two empirical facts organize the paper. First, the New Orleans public file changes abruptly on 28 July 2021. Before that date, no non-officer-initiated record has a valid arrival field without a dispatch field. Beginning on 28 July, that configuration appears and persists. At the same time, dispatch fields almost disappear from officer-initiated records even though their arrival fields remain nearly complete. Second, Hurricane Ida produces an additional, temporary reconfiguration within this new regime. On 31 August and 1 September, the public missing-dispatch share—the share missing a dispatch field among records with a valid arrival field—approaches one-half.

The Ida pattern is large relative to later ordinary weeks. Using every non-officer record in each half-day, the largest event-minus-prior-week change in a public field state is 53.2 percentage points. No post-change ordinary-week comparison is as large. A conditional multinomial bootstrap leaves Ida first in all 4,000 draws, although that exercise does not model time-series dependence. A composition-standardized comparison also ranks Ida first, but it retains incomplete common support and is therefore treated as secondary evidence.

The paper's contribution is about measurement, not about the causal effect of Ida. Public records show that the released measurement product changes before the hurricane and changes again during the event. They do not reveal whether the underlying cause was an operational workaround, an entry convention, a retention rule, or an export transformation. The implication is straightforward: a public CAD timestamp should not be treated automatically as an operational clock. Analysts must first establish the relevant denominator, field meaning, coverage, and lineage.

2 Related literature and contribution

A large economics and policing literature shows that response delays matter. Faster police arrival can improve crime clearance and reduce injuries, while traffic congestion, staffing, and queueing can slow first responders (Blanes i Vidal and Kirchmaier, 2018; DeAngelo et al., 2023; Brent and Beland, 2020; Mourtgos et al., 2024). These studies demonstrate the value of a valid response-time measure. The present paper asks the logically prior question: whether a public administrative export supplies a stable response clock in the first place.

The paper also relates to work on administrative measurement. Calls-for-service and police records are shaped by reporting, classification, discretion, verification, and data-processing rules (Klinger and Bridges, 1997; Boivin and Cordeau, 2011; Boivin and Ouellet, 2014; Simpson and Orosco, 2021). More generally, measurement error and misclassification can distort group comparisons and structural interpretation (Kapteyn and Ypma, 2007; Chen et al., 2011; Knox et al., 2020). The evidence here adds a time dimension: the joint presence of public CAD fields changes abruptly, so the mapping from activity to released data is not stable across the sample period.

Disaster and domestic-violence research provides substantive motivation. Hurricane exposure, displacement, psychological stress, partner conflict, and demand for services can move together (Anastario et al., 2009; Schumacher et al., 2010; Harville et al., 2011; First et al., 2022). Police composition can affect reporting (Miller and Segal, 2019), and pandemic-era studies show that calls, reports, and underlying victimization need not move one-for-one (Leslie and Wilson, 2020; Bullinger et al., 2021; Miller et al., 2022, 2024). Those studies do not validate a particular CAD timestamp. Ida is used here as a stress test of administrative observability, not as a treatment whose effect on domestic-violence incidence is estimated.

Finally, partial-identification methods provide the appropriate language when public records do not reveal a unique latent process (Manski, 1989, 1990, 2003; Tamer, 2010; Molinari, 2020). The paper does not propose a new general partial-identification estimator. It uses that discipline to distinguish quantities directly observed in the public file from quantities that are selected, bounded, or unavailable without additional evidence.

Relative to these literatures, the paper makes three contributions. It documents a date-localized change in the public CAD measurement regime; it shows that Ida produces an additional extreme but temporary reconfiguration within the later regime; and it translates those facts into practical requirements for measuring response time and reported-DV performance.

3 Data and public field measures

3.1 Official files and analysis population

The analysis uses the official New Orleans Calls for Service files for 2020–2026 (City of New Orleans, 2020, 2021, 2022, 2023, 2024, 2025, 2026). The reproduction package binds the 2020–2024 annual files to official dataset identifiers, row counts, observed date ranges, monthly source hashes, and deterministic annual bundle hashes.

The main analysis is restricted to records coded `SelfInitiated=N`, which the public data distinguish from items generated by officers in the field. These are referred to as *non-officer-initiated* records. Officer-initiated records are analyzed separately because their public dispatch-field convention changes sharply in July 2021.

For each record, define the dispatch field as present when `TimeDispatch` is nonblank. Define the arrival field as valid when `TimeArrive` parses and is no earlier than `TimeCreate`. These two indicators produce four public configurations.

Table 1. Four configurations in the released public record

Public configuration	Dispatch field	Valid arrival field	What the label does not establish
Neither field	No	No	No call activity or officer action.
Dispatch only	Yes	No	No physical arrival.
Arrival only	No	Yes	Physical arrival without dispatch.
Both fields	Yes	Yes	Stable endpoint meanings or a complete operational pathway.

These labels describe the released record. “Arrival only” does not mean that an officer physically arrived without being dispatched.

Three descriptive series summarize the public file: the share of records with a dispatch field, the share with a valid arrival field, and the *public missing-dispatch share*—among records with a valid arrival field, the share whose public dispatch field is missing. Calls are grouped by creation date; dispatch and arrival fields are read from the eventual released record.

4 A public-data regime change on 28 July 2021

The arrival-only configuration is completely absent from non-officer-initiated records before 28 July 2021. Across 419,840 records created from January 2020 through 27 July 2021, its count is exactly zero. It appears abruptly on 28 July and remains present thereafter.

A direct audit of the official July 2021 compressed CSV confirms that this is not an artifact of the aggregate pipeline. All nonblank dispatch and arrival fields in the audit interval parse under the stated rule, and the daily state counts agree exactly with the packaged aggregate tally. Table 2 shows the central transition.

The simultaneous movement across the two initiation streams is important. Officer-initiated records carry the public initiation label for items generated by officers in the field, yet their public

Table 2. The public-file transition, 27–29 July 2021

	27 July	28 July	29 July
Non-officer records	731	620	666
Non-officer arrival-only records	0 (0.0%)	34 (5.5%)	69 (10.4%)
Non-officer dispatch-field share	90.0%	82.1%	76.6%
Officer records	479	575	504
Officer dispatch-field share	100.0%	40.0%	1.0%
Officer valid-arrival share	100.0%	100.0%	100.0%

Percentages are calculated from the official daily source rows. Full state counts for 25–31 July are reported in the empirical supplement.

dispatch fields almost disappear while their arrival fields remain complete. The missing public dispatch field therefore cannot be read mechanically as the absence of an officer or of all operational activity. For this stream, the post-break change is necessarily a change in production of the released record, although the public evidence does not identify the specific entry, retention, transformation, or export step. The discrepancy lies somewhere between activity and released representation.

Figure 1 places the break in the longer series. Dispatch-field coverage among non-officer records is about 90 percent in 2020, then declines after July 2021 and reaches 65.4 percent in 2024. The arrival-only configuration, previously absent, becomes increasingly common.

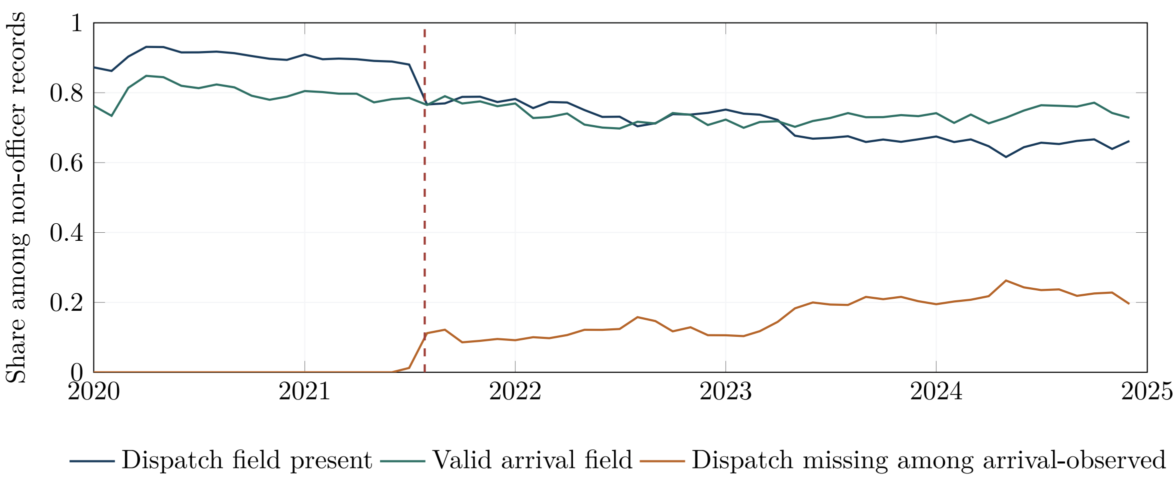

Figure 1. Public-field completeness among non-officer-initiated records. The dashed line marks 28 July 2021, the first date with an arrival-only record.

The later annual files confirm that this is a persistent measurement regime rather than a short Ida artifact. In the comparable non-officer denominator, dispatch-field coverage is 65.4 percent in 2024, 66.0 percent in 2025, and 66.8 percent in the 2026 snapshot. The much lower all-row rate of about 41.8 percent combines non-officer records with officer-initiated records, whose dispatch fields are almost always absent under the later convention.

Table 3. Dispatch-field coverage in the later public-data regime

Year	Non-officer records	Officer records	All public rows
2024	65.4%	1.1%	42.7%
2025	66.0%	0.9%	41.8%
2026 snapshot	66.8%	0.9%	41.9%

The all-row rate is not comparable to the non-officer series because it pools two streams with different recording conventions.

5 Hurricane Ida within the new regime

Hurricane Ida made landfall on 29 August 2021, five weeks after the public-file transition. Figure 2 shows the daily field pattern for non-officer records from one week before Ida through the following two weeks.

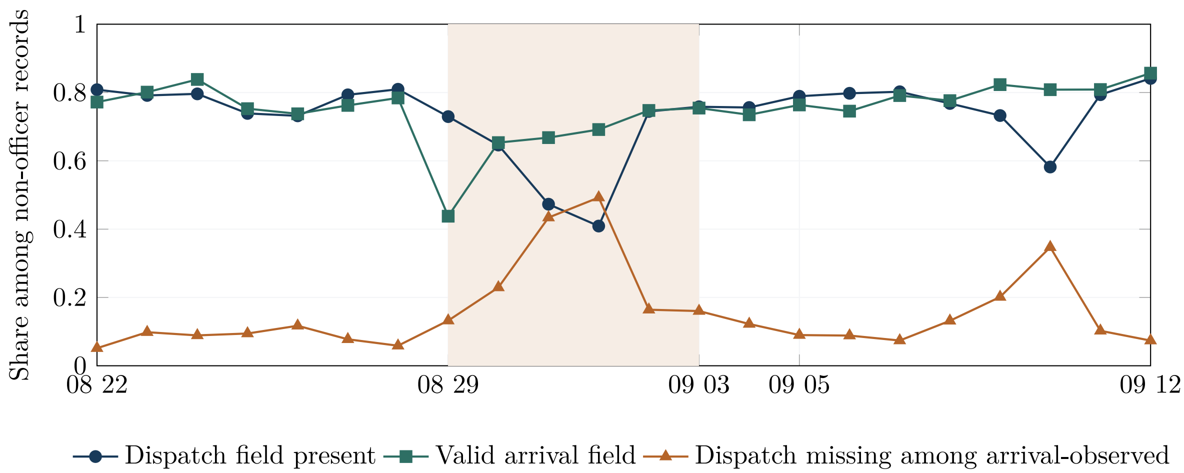

Figure 2. The public-field path around Hurricane Ida. Shading marks the five-day event window. The figure describes the released record, not physical dispatch or arrival.

During the preceding week, the public missing-dispatch share ranges from 5.1 to 11.7 percent. On landfall day, 29 August, the valid-arrival share falls to 43.8 percent from a preceding-week range of 74–84 percent, while dispatch-field coverage remains 72.9 percent. The public missing-dispatch share then rises to 22.9 percent on 30 August, 43.4 percent on 31 August, and 49.3 percent on 1 September as dispatch-field coverage falls to 47.3 and 40.9 percent on the latter two days. It falls to 16.4 percent on 2 September, returns to its pre-event range by 5 September, and shows one further one-day excursion to 34.7 percent on 10 September before settling. Thus the first event day is marked mainly by missing arrival fields, while the third and fourth days are marked by missing dispatch fields among arrival-observed records.

These movements are large, but their interpretation must remain administrative. The figures index calls by creation date and read fields from the eventual released record. They do not show a live system snapshot, nor do they establish that a unit physically arrived without dispatch. They show that records created during the event were ultimately released in a different field configuration.

6 How unusual was Ida?

6.1 Primary full-count comparison

For each five-day window, the analysis compares ten twelve-hour cells with the same half-days one week earlier. Within each cell it computes changes in three of the four public configurations: dispatch only, arrival only, and both fields present. The neither-field share is determined by the other three because the four shares sum to one. The summary statistic is the largest absolute change across the thirty cell-by-state comparisons:

$$M(w) = \max_{b,s} \left| \widehat{p}_{w,b,s} - \widehat{p}_{w-7,b,s} \right| . \tag{1}$$

For Ida, the largest public-state share changed by 53.2 percentage points ($M = 0.532$).

The primary reference comparison uses 151 ordinary windows for which both the event and baseline periods occur after 28 July 2021. Prespecified emergency and holiday windows remain excluded. This post-change set was defined after the July transition was identified, so it is a transparent descriptive sensitivity rather than a preregistered test.

The largest reference value is 0.310. Thus Ida is larger than all 151 post-change ordinary-week comparisons. A conditional window-wise multinomial bootstrap independently perturbs Ida and the reference windows. Ida ranks first in all 4,000 draws, with a 95-percent interval of $[0.475, 0.601]$ for its statistic. Because the procedure does not preserve serial or cross-window dependence, it is best read as a robustness check against ordinary categorical-count variation, not as formal time-series inference.

6.2 Secondary standardized comparison

A second estimator standardizes across common call-type, hour, and schema strata. It also ranks Ida first: the maximum standardized contrast is 0.507, compared with 0.328 for the largest of 153 prespecified references beginning on or after 1 July 2021. The supplement calls this the stage-era set and documents its inherited cutoff. However, the standardized analysis covers only 86.1 percent of event records and 77.0 percent of baseline records. Ida would fail the 0.90 support rule applied to candidate reference windows. The standardized result is therefore secondary; the full-count comparison is the main evidence.

Table 4. Ida relative to ordinary-week comparisons

Design	References	Ida largest change	Largest reference	Interpretation
Full-count, post-change	151	53.2 pp	31.0 pp	Primary descriptive comparison; uses every non-officer record in each half-day.
Standardized, stage-era	153	50.7 pp	32.8 pp	Same qualitative rank, but incomplete common support makes it secondary.

Two excluded episodes help interpret the result. The week after Ida has a large statistic of 0.478 because its baseline is Ida itself; it captures recovery rather than an independent shock. Hurricane Francine, a later-regime hurricane in September 2024, has a much smaller statistic of 0.110. Two pre-break windows later labelled as Tropical Storm Cristobal and its following comparison week have changes of 0.333 and 0.340, but those movements occur through the dispatch-only and both-field configurations rather than the arrival-only configuration. Ida is therefore not merely a generic consequence of comparing any hurricane week with the preceding week.

All other prespecified statistics and threshold variants are reported in the empirical supplement. They do not all rank Ida first. The main conclusion is narrow: Ida produces an unusually large change in the public field configuration.

7 What do the data suggest—and what remains unknown?

7.1 A recording-convention clue

The officer-initiated stream provides the most concrete clue. The public definition describes these items as generated by officers in the field. Before 28 July 2021, their dispatch and arrival fields are almost universally present. After the break, arrival remains nearly universal while dispatch falls to approximately 1 percent. The arrival-only configuration is therefore a normal released form for records carrying that public initiation label under the later regime.

During Ida, non-officer records move toward the same public configuration. One plausible pathway is that some calls normally represented with a dispatch field were entered, retained, or exported through a workflow that omitted it. That possibility is consistent with a change in how activity is represented in the released record, but it is not identified. Internal call-origin provenance, event-entry histories, unit-assignment histories, and export logs would be required to distinguish reclassification, operational workarounds, back-filling, and export omission.

7.2 Disposition mix is not the main accounting explanation

The late Ida movement also occurs within recorded final-disposition categories. On the afternoon and evening of 31 August, the public dispatch-field share among arrival-observed records falls by 49.5 percentage points. A change in the observed disposition mix would have increased the share by about 1.2 points; within-disposition changes account for approximately -50.7 points. The following morning produces nearly the same decomposition. The observed final-disposition mix therefore does not explain the aggregate decline.

This is an accounting result, not a mechanism test. The analysis conditions on records with arrival fields and uses final recorded dispositions. Selection into that population, changes within broad categories, and changes in recording practice remain possible.

7.3 What public documents establish

Official metadata define the public field labels and identify the data provider. The DataNOLA page also warns that records are preliminary, may contain mechanical or human error, may change after investigation, and should not be used mechanically for comparison over time (City of New Orleans, 2021). Public oversight materials document the Ida emergency context (Office of the Independent Police Monitor, 2021), while NOPD policy materials describe intended communications and records practices (New Orleans Police Department, 2023).

The public chronology rules out two proposed technology explanations. The publicly indexed Carbyne APEX cutover occurred in June 2022, eleven months after the July 2021 change (Orleans Parish Communication District, 2022). The Hexagon OnCall Records project was contracted but never launched and therefore cannot explain the transition (New Orleans Office of Inspector General, 2024). These exclusions are useful, but they do not identify the actual cause.

No public versioned changelog or internal-to-public reconciliation was available for this analysis, and no agency confirmation of the July change was obtained. A focused public-records request to OPCD for the CAD export configuration and change history covering July 2021 is the most direct next step for resolving the mechanism. Until such evidence is available, the strongest supported conclusion is a discontinuity in the released record, not a named institutional mechanism.

8 Implications for response-time and reported-DV research

The paper's measurement lesson applies beyond Hurricane Ida. Public CAD can support a response-time or reported-DV measure only when the target population and the public endpoints are explicit.

Table 5. What the public file supports

Target	What the public file directly supports	Additional evidence needed
Field completeness	Shares of records with each released public field, on a declared denominator.	None beyond source and parsing documentation.
Response time	A duration among records with both declared endpoints. This is a selected-record statistic when endpoints are missing.	Validated endpoint meanings, coverage on the full denominator, and evidence that the fields map to the intended operational events.
Queueing or capacity	No direct measure from endpoint presence alone.	Calls holding, queue flows, unit availability, assignments, and internal event histories.
Reported-DV performance	Counts under a declared classification rule and public-field completeness within that classified population.	Validated classification, treatment of unresolved records, endpoint coverage, and lineage to reports or follow-up stages.

Reported CAD calls are not equivalent to underlying domestic-violence incidence or complete help-seeking. A future DV analysis should preserve unresolved classifications rather than force every record into or out of the target group, and it should report endpoint coverage before interpreting observed durations. Ida motivates this measurement work, but the current system-level analysis does not estimate a DV effect.

9 Limitations

The analysis uses eventual public records rather than internal event histories. The July change is verified in the official source bytes, but its institutional cause is not observed. The post-change comparison set was defined after the break was identified and excludes specified emergency and holiday contexts. Reference windows overlap; retaining every second window removes adjacent overlap and leaves Ida first in both alternating phases (supplement). The bootstrap does not preserve all time-series dependence. Public `SelfInitiated=N` is not equivalent to "911 only." Weather, holidays, mobility, evacuation, reporting behavior, call composition, suppression, and other contemporaneous conditions can affect the observed record.

These limitations restrict interpretation but do not erase the descriptive result. The released field configuration changes abruptly in July 2021, changes further during Ida, and follows different conventions across initiation streams.

10 Conclusion

Public CAD files are valuable administrative records, but their timestamps are not operational ground truth by definition. In New Orleans, a new arrival-only configuration appears suddenly on 28 July 2021, accompanied by an almost complete disappearance of dispatch fields from officer-initiated records. Five weeks later, Hurricane Ida produces an additional two-day reconfiguration that is larger than every post-change ordinary-week comparison.

The public evidence identifies the timing and magnitude of changes in the released record. It does not identify the underlying operational sequence or the institutional mechanism that produced those fields. Analysts should therefore treat public CAD timestamps as administrative measures until their meanings, coverage, and lineage have been validated for the specific performance question.

Data and code availability

Replication materials, source bindings, the empirical supplement, and the release verifier are available in the public repository at https://github.com/yuwen3756-hue/public-cad-measurement-police-responsiveness-hurricane-ida. The repository provides derived aggregate artifacts and reproducible code; official raw CAD files remain identified by first-hand public-source links and hashes in the reproduction materials.

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Online Empirical and Methods Supplement

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Scope

This supplement follows the main paper's empirical sequence: source lineage and parsing; the July transition; initiation-stream and current-denominator evidence; the Ida daily path; the primary full-count comparison; then standardized, bootstrap, secondary-statistic, institutional, and reported-DV details. Extended identified-set proofs, constrained-program certificates, and formal-verification material are kept in the separate *Legacy Mathematical and Formal-Verification Archive* and are not appended to the professor-facing paper.

Contents

1 Source lineage and parsing rules

The analysis distinguishes four evidence classes.

1. **First-hand public bytes:** official DataNOLA CSVs and their hashes.
2. **Deterministic aggregate transforms:** daily, half-day, monthly, and annual state counts derived from those files.
3. **Institutional documents:** official metadata, data dictionary, policy, oversight, and audit reports that define labels or exclude candidate explanations.
4. **Latent operational objects:** internal dispatch events, queue states, availability, entry histories, overwrite logs, and export transformations, which are not observed.

The first three classes identify a public-record regime change and delimit candidate interpretations. They do not turn the fourth class into observed data. Accordingly, J_{01} is always a public configuration—arrival field present, dispatch field absent—rather than a claim of physical arrival without dispatch.

1.1 Annual public-source lineage

1.1.1 Dataset binding

Table 1 binds each analysis year to the official DataNOLA identifier and the first-hand cache audit. Each annual bundle hash is computed by sorting the twelve monthly filenames and hashing the filename plus each file's SHA-256 digest. This avoids concatenating or redistributing the raw source files. The monthly cache did not retain a reliable acquisition-date field, so the table reports that limitation rather than converting filesystem timestamps into retrieval dates. All sources were reverified on 25 August 2026.

Table 1. Annual DataNOLA source lineage, 2020–2024

Year	Dataset	Observed creation range	Rows	Files	Annual bundle SHA-256
2020	hp7u-i9hf	1 Jan.–31 Dec. 2020	432,892	12	`a1d3d7fffb6a34e7...da7cc0db1`
2021	3pha-hum9	1 Jan.–31 Dec. 2021	428,315	12	`53b0ae39abbbc363...a7edc559`
2022	nci8-thrr	1 Jan.–31 Dec. 2022	354,207	12	`1ca797cd30109a80...e3d1e4d8`
2023	pc5d-tvaw	1 Jan.–31 Dec. 2023	325,091	12	`68d583fe0c4dbb43...4f011a13a`
2024	2zcj-b6ts	1 Jan.–31 Dec. 2024	327,696	12	`041f7ac52458c797...b4f6f464`

Exact observed minimum and maximum timestamps, full hashes, monthly-file hashes, byte counts, local cache mtimes, official URLs, parser version, and verification date are in `r15_annual_source_lineage.csv` and `r15_public_source_audit.json`. Retrieval dates were not recorded in the monthly cache.

The annual row counts are raw public-file rows before the paper's initiation restriction. Analysis denominators are constructed after parsing `SelfInitiated`. No narratives or addresses are used. Item identifiers are used only by the upstream canonical pipeline for deduplication and are not written to review artifacts.

1.1.2 Official metadata and changelog

The official 2021 page identifies OPCD as data provider, reports 21 public columns, supplies a data-dictionary attachment, and warns about preliminary classifications, later changes, mechanical or human error, sequencing, and over-time comparisons (City of New Orleans, 2021). When inspected, its public "Dataset Changelog" stated that no changes had been archived. This is itself an observability limitation: the public page does not provide a dated change record that could explain 28 July 2021.

2 July transition: raw audit and aggregate parity

2.1 Parsing and state construction

The audit applies the same public-field definitions as the aggregate analysis:

$$D_i = \mathbf{1}\{\texttt{TimeDispatch}_i \text{ is nonblank}\}, \tag{1}$$

$$A_i = \mathbf{1}\{\texttt{TimeArrive}_i \text{ parses and is no earlier than } \texttt{TimeCreate}_i\}. \tag{2}$$

Timestamps are parsed as local ISO strings with a `T` or single-space separator. Empty strings and the textual values `null`, `none`, `nan`, and `nat` are missing sentinels. All nonblank dispatch and arrival values in the audit interval parse. Format-class counts show no switch that could produce the break mechanically.

Table 2. Complete aggregate audit, 25–31 July 2021

Date	Stream	N	D nonblank	D parsed	A nonblank	A valid	J_{00}	J_{10}	J_{01}	J_{11}
25 Jul.	Non-officer	721	631	631	532	532	90	99	0	532
26 Jul.	Non-officer	733	663	663	565	565	70	98	0	565
27 Jul.	Non-officer	731	658	658	585	585	73	73	0	585
28 Jul.	Non-officer	620	509	509	502	502	77	41	34	468
29 Jul.	Non-officer	666	510	510	532	532	87	47	69	463
30 Jul.	Non-officer	705	542	542	559	559	98	48	65	494
31 Jul.	Non-officer	725	584	584	572	572	96	57	45	527
25 Jul.	Officer	377	377	377	377	377	0	0	0	377
26 Jul.	Officer	404	404	404	404	404	0	0	0	404
27 Jul.	Officer	479	479	479	479	479	0	0	0	479
28 Jul.	Officer	575	230	230	575	575	0	0	345	230
29 Jul.	Officer	504	5	5	504	504	0	0	499	5
30 Jul.	Officer	548	3	3	548	548	0	0	545	3
31 Jul.	Officer	389	1	1	389	389	0	0	388	1

2.2 What the audit resolves

The audit resolves three concerns. The first J_{01} date is present in official source bytes. The change survives a parseability audit. And the abrupt officer-stream shift shows that the non-officer pattern is part of a broader public-record transition rather than an isolated fluctuation in one classified group.

The independent raw-to-aggregate comparison checks fourteen date-by-stream rows and all four public states. All 56 state cells match exactly. This establishes that the July result in the aggregate series is a lossless transform of the audited source counts.

The audit does not resolve the institutional cause. It has no event-entry history, user action, intermediate CAD object, export log, or versioned retention rule. The strongest supported wording is therefore "public-recording transition associated with initiation status," not a named software or operational mechanism.

3 Ida path and maximum-change statistics

The daily file path begins with the intuitive quantities used in the main paper. The public missing-dispatch share—among non-officer records with a valid arrival field, the share whose public dispatch field is missing—rises from 5–12 percent in the preceding week to 43.4 percent on 31 August and 49.3 percent on 1 September. On landfall day, 29 August, the valid-arrival share instead falls to 43.8 percent while dispatch-field coverage remains 72.9 percent. The machine-readable daily series contains 22 dates from 22 August through 12 September 2021; it also records the separate one-day missing-dispatch excursion to 34.7 percent on 10 September. The half-day comparison uses ten twelve-hour event cells and the corresponding cells seven days earlier.

For the primary full-count design, let $\widetilde{p}_{w,b,s}$ be the unconditional share in state s for half-day b of window w. The largest percentage-point change is

$$M^{\text{full}}(w) = \max_{b,s} \left|\widetilde{p}_{w,b,s} - \widetilde{p}_{w-7,b,s}\right|. \tag{3}$$

It uses every non-officer record in each cell. Ida's value is 0.5320. The post-change comparison contains 151 ordinary reference windows for which both the event and baseline sides begin after 28 July 2021; its largest reference value is 0.3098. This reference set was defined after the July transition was identified and is therefore a descriptive post-hoc comparison.

For half-day bin t, state $j \in \{J_{01}, J_{10}, J_{11}\}$, and common stratum x, let p^E_{tjx} and p^B_{tjx} be event and baseline shares and let w^B_{tx} be the baseline weight over common support. The standardized contrast is

$$G_{tj} = \sum_{x \in \mathcal{X}^E_t \cap \mathcal{X}^B_t} w^B_{tx}(p^E_{tjx} - p^B_{tjx}). \tag{4}$$

The standardized amplitude is $M_{\max} = \max_{t,j} |G_{tj}|$. The event window begins 29 August 2021 and contains ten twelve-hour bins; the baseline is seven days earlier.

The standardized stage-era universe consists of 153 retained ordinary-week references beginning on or after 1 July 2021. Sixty-six eligible full-universe references occur entirely before 28 July, when J_{01} is structurally absent, and 64 start before the 1 July stage cutoff. These facts explain why a common formula does not imply a common measurement regime.

Two additional full-count reference sets are reported for transparency:

- the inherited stage-era membership set, which removes support selection but retains the existing membership rule; and
- a post-change set requiring both event and baseline sides to begin after 28 July, while retaining the frozen context exclusions.

The second is the post-change comparison used as the primary descriptive reference in the main paper.

Table 3. Ida common-support coverage and record counts

Bin	Event N	Baseline N	Event coverage	Baseline coverage
B1	276	244	0.928	0.754
B2	674	335	0.901	0.612
B3	206	291	0.748	0.505
B4	495	433	0.919	0.815
B5	295	290	0.766	0.838
B6	464	421	0.873	0.812
B7	311	268	0.846	0.802
B8	464	379	0.851	0.836
B9	240	255	0.829	0.812
B10	453	416	0.834	0.851

4 Secondary standardized estimator and common-support audit

The minimum coverages are 0.748 and 0.505; record-weighted coverages are 0.861 and 0.770. The symmetric 0.90 rule fails. Consequently, the standardized result cannot be described as the full event-window population. The full-count statistic uses all 3,878 event and 3,332 baseline records.

5 Bootstrap results and dependence caveats

5.1 Standardized windows

Within each observed stratum, records are resampled as multinomial draws over $(J_{00}, J_{10}, J_{01}, J_{11})$ while holding the observed stratum totals and baseline weights fixed. This is exactly the record-level nonparametric bootstrap available from the aggregate state counts. It captures record sampling conditional on the observed strata; it does not capture serial dependence, regime-boundary uncertainty, or design selection. Adjacent references share a week because one event period becomes the next baseline, so window-wise resampling does not preserve their cross-window dependence.

5.2 Full-count windows and rank distribution

The same multinomial logic is applied to unconditional half-day state counts. Ida's full-count interval is $[0.4754, 0.6015]$. The strongest post-change reference point is 0.3098 with interval $[0.2599, 0.3799]$; the remaining top-five intervals have upper endpoints between 0.3636 and 0.3843. In each of 4,000 conditional window-wise draws, Ida is ranked against all 151 separately perturbed post-change references. Ida ranks first in every draw, but the exercise does not model temporal dependence and is not formal rank inference for an exchangeable time series.

An alternating-window sensitivity retains every second chronological post-change reference, which removes the direct event-to-next-baseline overlap. Ida is rank 1/77 including Ida in one phase and 1/76 in the other. The two largest retained reference values are 0.298 and 0.310. This does not eliminate longer-run dependence; it reports exactly how the result changes when adjacent overlap is removed.

Table 4. Bootstrap intervals for Ida and the five strongest standardized references

Window	Point	2.5th percentile	97.5th percentile
Ida, 29 Aug. 2021	0.5072	0.4597	0.5656
15 May 2022	0.3282	0.2655	0.3954
22 May 2022	0.3083	0.2528	0.3725
12 May 2024	0.3064	0.2502	0.3880
5 May 2024	0.2904	0.2355	0.3660
10 Dec. 2023	0.2656	0.1630	0.3764

Table 5. Full-count bootstrap: Ida and five strongest post-change references

Window	Point	2.5th percentile	97.5th percentile
Ida, 29 Aug. 2021	0.5320	0.4754	0.6015
15 May 2022	0.3098	0.2599	0.3799
10 Dec. 2023	0.2978	0.2140	0.3843
19 Jun. 2022	0.2959	0.2384	0.3636
5 May 2024	0.2941	0.2501	0.3735
12 May 2024	0.2926	0.2238	0.3773

6 Secondary statistics, LP, thresholds, and timing placebo

The 95th-percentile maximum-cell thresholds are 0.2094 in the full set, 0.2247 in the stage-era set, and 0.2057 in the same-season set. They yield 9, 8, and 9 exceeding cells. Reporting one threshold without the other two would overstate the stability of a binary label.

The constrained LP is a robustness calculation. Its Ida lower endpoint exceeds 0.25 by 0.000734, while 50 of 217 reference intervals have width above 0.01 and the maximum width is 0.070. The numerical LP status is therefore a tested numerical result, not an exact rational certificate. Formal code verifies arithmetic and finite certificate logic; it does not verify the empirical meaning of public fields.

The shifted-timing exercise is unfavorable to an institutional-timing interpretation: zero-padded translations of the support library fit better than the documented alignment. This does not alter the public-field contrast, but it means that the timing calculation does not identify a mechanism.

Table 6. Secondary stage-era statistics

Statistic	Ida	Largest reference	Rank including Ida
M_{max} / A	0.5072	0.3282	1/154
σ_1	1.0837	0.6011	1/154
σ_2	0.2623	0.2451	1/154
σ_3	0.0858	0.1112	11/154
U_{direct}	0.4223	0.3282	1/154
U_{full} upper	0.2507	0.2125	1/154
Q upper	0.4944	0.9408	100/154
D upper	0.1716	0.2835	4/154

7 Excluded episodes and recovery comparisons

The frozen exclusion list omitted Tropical Storm Cristobal, which made Louisiana landfall on 7 June 2020 (National Hurricane Center, 2021). The 7 and 14 June windows therefore remain ordinary members of the full-qualified rank; the second uses the Cristobal week as its baseline. Their large pre-break full-count movements occur through the available J_{11}/J_{10} states, not through J_{01}. This post-hoc label documents an incomplete exclusion list and is not used to alter membership. The other labelled episodes in the table are excluded from the ordinary-week rank. The large post-Ida statistic compares a recovery week with Ida itself, while Francine is much smaller in the later regime. These comparisons reinforce the need to distinguish event stress from the measurement object used to observe it.

Table 7. Labelled emergency and disturbance windows: full-count maximum-cell contrast

Episode window	Event records	Baseline records	Raw $M_{\max}$
Hurricane Laura, 23 Aug. 2020	3,316	3,515	0.084
Hurricane Zeta, 25 Oct. 2020	3,963	3,638	0.129
Tropical Storm Cristobal, 7 Jun. 2020	3,431	3,323	0.333
Following comparison week, 14 Jun. 2020	3,454	3,431	0.340
February freeze, 14 Feb. 2021	2,851	3,329	0.113
Hurricane Ida, 29 Aug. 2021	3,878	3,332	0.532
Post-Ida week, 5 Sep. 2021	3,827	3,878	0.478
Hurricane Francine, 8 Sep. 2024	2,886	2,882	0.110

8 Initiation-stream and current-denominator evidence

The main analysis excludes officer-initiated records because their public dispatch convention differs radically. Their pattern is nevertheless analytically valuable. It shows that J_{01} is a normal public configuration in one stream after the break and that pooling initiation streams creates a denominator artifact. It does not prove that non-officer calls were operationally converted into officer activity.

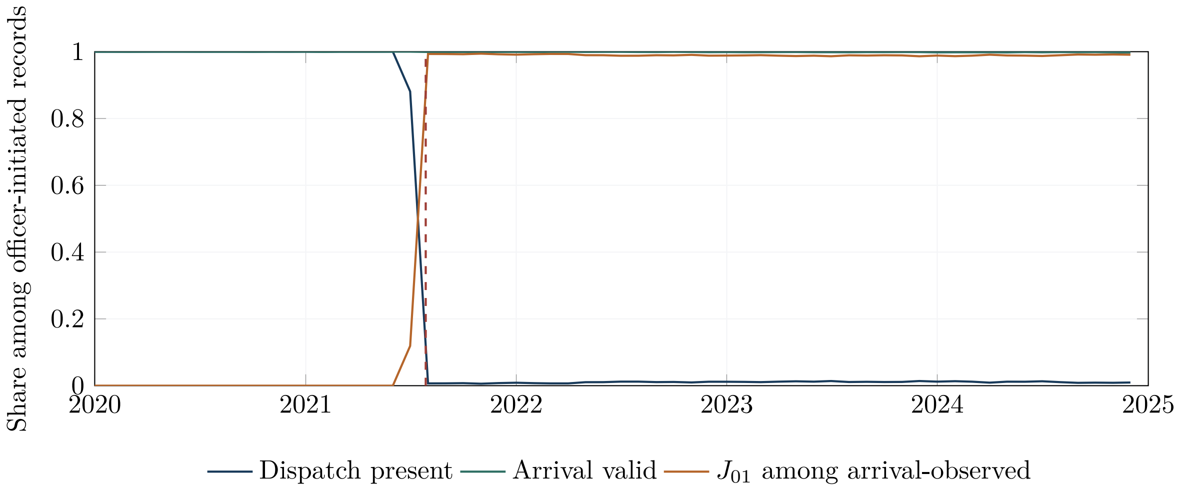

Figure 1. Officer-initiated public-field pattern, 2020–2024. Dispatch presence moves from essentially complete to approximately 1 percent while arrival remains essentially complete.

Table 8. Annual field completeness by initiation stream

Year	Stream	Records	Dispatch share	Arrival share
2020	Non-officer	271,868	0.903	0.801
2021	Non-officer	258,160	0.845	0.783
2022	Non-officer	243,179	0.745	0.724
2023	Non-officer	225,389	0.693	0.722
2024	Non-officer	212,314	0.654	0.742
2020	Officer	161,024	1.000	1.000
2021	Officer	170,155	0.617	1.000
2022	Officer	111,028	0.010	0.999
2023	Officer	99,702	0.012	0.999
2024	Officer	115,382	0.011	0.998

9 Disposition accounting

Among arrival-observed records, let p_{dt} be the recorded-disposition share and q_{dt} the dispatch-field rate within disposition d at time t. The aggregate change decomposes exactly as

$$\Delta\bar{q} = \sum_d \bar{p}_d \Delta q_d + \sum_d \bar{q}_d \Delta p_d, \tag{5}$$

where bars denote midpoints. For the two largest late Ida contrasts, the totals -0.4950 and -0.4824 decompose into within-disposition terms -0.5069 and -0.4946 plus composition terms $+0.0119$ and $+0.0123$. The observed final-disposition mix is not sufficient to explain the shift. Because the sample conditions on arrival observation and uses final recorded dispositions, the decomposition does not identify operational routing or causal mechanisms.

Table 9. 2025 and 2026 snapshot field completeness

Year	Denominator	Records	Dispatch present	Dispatch share
2025	Non-officer	207,050	136,712	0.6603
2025	Officer	122,720	1,117	0.0091
2025	All rows	329,770	137,829	0.4180
2026 snapshot	Non-officer	130,346	87,129	0.6684
2026 snapshot	Officer	79,483	738	0.0093
2026 snapshot	All rows	209,829	87,867	0.4188

9.1 Current-data denominator audit

The 2025 all-row rate is arithmetically correct but scientifically noncomparable to the non-officer series. The comparable series is 0.6538 (2024), 0.6603 (2025), and 0.6684 (2026 snapshot). The overall rate is low because officer-initiated records have nearly universal arrival fields and almost no dispatch fields under the later public convention.

10 Institutional-source audit

Table 10. Public institutional documents and what they identify

Source	Public status	What it adds	What it does not add
DataNOLA 2021 page and dictionary	Official page and attachment public	Field labels, provider, schema, preliminary-data warning, no archived changelog	Entry history, overwrite rule, internal/public lineage, July change cause
OIPM Hurricane Ida report	Official PDF public	Oversight context, emergency conditions, monitored activities	Machine-event semantics, export history, causal explanation of J states
NOPD manual chapters	Public editions located	Intended communications, call-response, and records practices	Complete exact-edition set for July–September 2021; proof of implementation
OPCD Carbyne APEX notice	Official page indexed; current URL returns not found	Cutover date of 24 June 2022	Explanation of July 2021 break
OIG Hexagon report	Official PDF public	OnCall Records project history; system was not launched	Evidence of a CAD-export transition

These documents supply genuine missing pieces: source semantics, public warnings, chronology, and negative evidence against two named technology stories. They still leave the decisive bridge missing. A mechanism claim would require a versioned changelog or internal/public reconciliation showing how a particular event and entry history became the released row.

11 Reported-DV measurement implications

The paper conditions on reported calls created in CAD. It does not observe latent need, attempts to seek help, contacts not converted into CAD items, or underlying domestic-violence incidence.

Broad text or code rules can also create false positives, and public suppression can create false negatives. The recommended classification remains $1/0/U$ until an authoritative crosswalk resolves the uncertain group.

If n_1, n_0, and n_U are the definitely included, definitely excluded, and unresolved counts, then the sharp no-assumption prevalence bounds are

$$\frac{n_1}{n_1 + n_0 + n_U} \le p \le \frac{n_1 + n_U}{n_1 + n_0 + n_U}. \tag{6}$$

For response-time outcomes, endpoint coverage must be stated on the full declared denominator. A duration among rows with both timestamps is a selected-stage statistic unless missing endpoints are bounded or independently qualified. The system-level Ida stress test shows why these requirements are substantive rather than cosmetic.

12 Reproduction and machine-consistency contract

The package build performs the following sequence:

1. rebuild aggregate diagnostics and all tables from the packaged frozen tally and reference artifacts;
2. optionally rerun the raw-source lineage audit against the canonical official cache;
3. compile the main paper, this supplement, research status note, and separate legacy archive;
4. combine only the main paper and empirical supplement into the professor-facing PDF;
5. check exact headline values, rank denominators, support status, threshold counts, 2025/2026 denominator arithmetic, source-audit invariants, and forbidden revision-history language in the main source;
6. verify PDF names, page relationships, text presence, manifest hashes, and the frozen reproduction snapshot.

The ordinary builder uses Python's standard library and NumPy. SciPy is required only by the inherited constrained-optimization reproduction path, and pypdf is used to combine PDFs. Pandas is not required. The package manifest covers all curated files except the manifest itself and any generated zip. Text normalization performed by packaging is checked through the rebuilt numerical and PDF parity tests rather than treated as byte identity with an external working tree.

13 Interpretive checklist

Before using a public CAD duration as a performance measure, report:

1. the source identifier, observed date range, row count, and source hash;
2. the declared denominator and initiation-stream rule;
3. the exact start and endpoint semantics;
4. field nonblank, parseability, and ordering rates;
5. missing-endpoint handling and support coverage;
6. versioned institutional lineage, if available;
7. whether the target is a released-field statistic, selected-stage measure, bounded performance functional, or unavailable latent object;
8. shock exclusions, day-of-week, holidays, heat, weather, seasonality, COVID-era behavior, re-

porting behavior, help-seeking, suppression, and classification uncertainty.

References

City of New Orleans (2021). Calls for service 2021. Data.NOLA dataset 3pha-hum9; official page, disclaimer, attached data dictionary, and empty archived changelog verified 25 August 2026.

National Hurricane Center (2021). Tropical cyclone report: Tropical storm cristobal (al032020). Official report documents Louisiana landfall at 2200 UTC on 7 June 2020; verified 25 August 2026.

14 Numerical claim index

Table 11 identifies the package object that supplies each headline number. The release verifier recomputes the claims from these machine-readable objects; the table is an orientation aid, not an independent evidentiary source.

Table 11. Headline claims and their machine-readable sources

Claim	Reported value	Primary package object
Standardized Ida maximum	0.5072; rank 1/154	`r15_aggregate_diagnostics.json`
Full-count stage-era result	0.5320; rank 1/151 including Ida	`r15_aggregate_diagnostics.json`
Post-hoc post-change result	0.5320; rank 1/152 including Ida	`r15_aggregate_diagnostics.json`
Alternating non-overlap sensitivity	ranks 1/77 and 1/76 including Ida	`r15_1_nonoverlap_sensitivity.csv`
Threshold exceedances	9, 8, and 9	`r15_aggregate_diagnostics.json`
Standardized bootstrap interval	[0.4597, 0.5656]	`r15_bootstrap_window_maxima.csv`
Full-count bootstrap interval	[0.4754, 0.6015]	`r15_bootstrap_window_maxima.csv`
2025 non-officer dispatch share	136,712/207,050 = 0.6603	`r15_current_denominator_audit.csv`
2025 officer dispatch share	1,117/122,720 = 0.0091	`r15_current_denominator_audit.csv`
First non-officer J_{01} day	28 July 2021	`r15_raw_july_25_31_audit.csv`
Raw/aggregate July state parity	56/56 state cells match	`r15_1_raw_aggregate_parity.csv`
Annual public-source bindings	five official dataset IDs	`r15_annual_source_lineage.csv`

The aggregate objects are fully contained in the reproduction snapshot. The raw July and annual-lineage objects add first-hand source checks against the canonical public cache. Institutional documents remain a separate evidence class: they inform field semantics, warnings, and chronology, but do not generate or alter the numerical statistics above.